Lecture 4 - 01/22/20

Last time

For a rank-N matrix $A \in \mathbb{C}^{M \times N}$, $\exists A^L \in \mathbb{C}^{N \times M}$ s.t

① $A(A^L \underline{b}) = \underline{b}$, $\forall \underline{b} \in R(A)$

② $A^L(A\underline{x}) = \underline{x}$, for $\forall \underline{x} \in \mathbb{C}^N$

Ex: $A = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $A^L = [a_1 \ a_2]$ $\Rightarrow A^L A = 1 \Rightarrow a_1 + 2a_2 = 1 \Rightarrow a_1 = 1 - 2a_2$

Degree of freedom

$\Rightarrow$ We have many infinitely many options for A^L

① $[1, 0]$ ② $[0, 1/2]$ ③ $[-1, 1]$

Consider: $\underline{b} = \begin{bmatrix} 3 \\ 6 \end{bmatrix} \in R(A)$ $\rightarrow A^L \underline{b} =$ ① 3 ② 3 ③ 3

$\underline{x} = 3$

Consider: $\underline{b} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \notin R(A)$ $\rightarrow A^L \underline{b}$ ① 3 ② 2 ③ 1

Properties

- AA^L is a Left identity of A $\rightarrow AA^L A$
- AA^L is Idempotent: $(AA^L)^2 = (AA^L AA^L) = AA^L$
- The left inverse is unique iff A is square and nonsingular.

Proof (third Property)

$$A^L A = I_N \Rightarrow A^H (A^L)^H = I_N \Rightarrow A^L = \begin{bmatrix} \underline{a}_1^H \\ \vdots \\ \underline{a}_N^H \end{bmatrix} \Rightarrow A^H \underline{a}_K = \underline{e}_n \quad \text{for } n = 1, \dots, N$$

Uniqueness of $\underline{a}_n \Rightarrow \mathcal{N}(A^H) = \{0\} \Leftrightarrow A$ has linear independent rows $\Rightarrow M \leq N$

but we have A full column rank $\Rightarrow M \geq N \Rightarrow M = N \Rightarrow A$ is square and singular

Right inverses (A has full row rank)

Theorem (Existence of a linear right inverse)

If $R(A) = \mathbb{C}^M$, $\exists A^R \in \mathbb{C}^{N \times M}$ s.t $AA^R \underline{b} = b \quad \forall \underline{b} \in \mathbb{C}^M$

Proof for each $\underline{b} \in \mathbb{C}^M$ exists at least one $\underline{x} \in \mathbb{C}^N$ s.t $A\underline{x} = \underline{b}$

Let $\underline{x}_m$ be any solution to $A\underline{x} = \underline{e}_m$ and Let $A^R = [\underline{x}_1, \underline{x}_2, \dots, \underline{x}_M]$

Let $\underline{b} = \sum_{m=1}^{M} b_m \underline{e}_m \Rightarrow A(A^R \underline{b}) = \sum_{m=1}^{M} b_m AA^R \underline{e}_m = \sum_{m=1}^{M} b_m A\underline{x}_m = \sum b_m \underline{e}_m = \underline{b}$

Note: $AA^R = I_M$

Properties

- A^R is a right inverse of A iff $(A^R)^H$ is a left inverse for A^H
- $A^R A$ is a right identity for A; $AA^R A = A$
- AA^R is idempotent
- A^R is unique if and only if A is square and non singular.

What if solution doesn't exist?

What to do when $A\underline{x} = \underline{b}$ can never be satisfied for any $\underline{x}$?

$\Rightarrow$ Relax the requirement to $A\underline{x} \approx b$

$$\Rightarrow \hat{x} = \underset{\underline{x} \in \mathbb{C}^N}{\operatorname{argmin}} \, d(A\underline{x}, \underline{b})$$

Some kind of distance

(Why not actual distance?)

Often times the choice of $d(\cdot,\cdot)$ is statistically motivated.

for now, $\hat{\underline{x}} = \underset{\underline{x} \in \mathbb{C}^N}{\operatorname{argmin}} \| A\underline{x} - \underline{b} \|$ norm

Norms

$\|x\|$ $\quad \|\cdot\| : \mathcal{X} \to \mathbb{R}$ which satisfies

① $\|\underset{\sim}{x}\| \geq 0 \quad \forall x \in \mathcal{X}$ and $\|x\| = 0$ iff $x = 0$ (Positivity)

② $\|\underline{x} + \underline{y}\| \leq \|x\| + \|y\|$ (triangle inequality

③ $\|\alpha x\| = |\alpha| \, \|x\|$ (Homogeneity Property)

Example • Vector spaces $\mathbb{C}^N$ and $\mathbb{R}^N$

- $\ell_p(\mathbb{C}^N)$ (ℓ_p-norm) $p \geq 1 \to \|\underline{x}\|_{\ell_p} \triangleq \left(\sum_{n=1}^{N} |x_n|^p\right)^{1/p}$

for $p=2$: - Euclidean norm

- Hilbert space norm
- least squares filter design
- Parseval's theorem
- $\|\underline{x}\|_{\ell_2}^2 \Rightarrow$ negative log-likelihood for zero mean Gaussian noise (iid)

for $p=1$: the ℓ_1-norm

- sparsity
- Absolute summability
- Uniform Convergence of the DTFT

for $\lim_{p \to \infty}$ "the ℓ_∞-norm" $\|\underline{x}\|_\infty = \max_n |x_n|$

- Chebyshev filter design
- BIBO stability : Bounded signals $\to$ finite ℓ_∞ norm

• Vector spaces of finite sequences

$\ell_p(\mathbb{Z})$ $\quad \|\underline{x}\|_{\ell_p} = \left(\sum_{n \in \mathbb{Z}} |x_n|^p\right)^{1/p}$

Notation

The maximum may not exist!

$\|\underline{x}\|_{\ell_\infty} = \sup_n |x_n|$

• Vector space of Continuous-time signals

$\mathcal{L}_p(\mathbb{R})$ $\quad \|\underline{x}\|_{\mathcal{L}_p} \triangleq \left[\int_{\mathbb{R}} |x(t)|^p dt\right]^{1/p}$

- Can be generalized $\mathcal{L}_p(\mathbb{R}^2)$ - Images v $\mathcal{L}_p(\mathbb{R}^3)$ - Videos

For now, restrict atention to $\mathbb{C}^N$ and the ℓ_2-norm

Consider: $\hat{\underline{x}} = \underset{\underline{x} \in S}{\operatorname{argmin}} \|\underline{x} - \underline{b}\|_{\ell_2}$

<u>ex</u>: Let $S = \{s \in \mathbb{R} : 0 \le s < 1\}$

$\hat{\underline{x}} = \underset{\underline{x} \in S}{\operatorname{argmin}} |x - b|$ $\Rightarrow$ $b = -1 \rightarrow \hat{x} = 0$

$b = 0.4 \rightarrow \hat{x} = 0.4$

$b \geq 1 \rightarrow$ there is no solution

$\Rightarrow$ Optimization problems don't always have solutions

We will focus on $\hat{\underline{x}} = \underset{\underline{x} \in \mathbb{C}^N}{\operatorname{argmin}} \|A\underline{x} - \underline{b}\|_{\ell_2}^2$ (linear least-squares)

$\hat{\underline{x}} = \underset{\underline{x} \in S}{\operatorname{argmin}} \|\underline{x} - \underline{b}\|_{\ell_2}^2$ (eg., $S = \mathcal{R}(A)$)

<u>Orthogonality</u> $\rightarrow$ Inner product

<u>Def</u> for $\underline{x}, \underline{y} \in \mathbb{C}^N$, the inner product is $\underline{y}^H \underline{x} = \sum_{n=1}^{N} y_n^* x_n$

Note $x^H x = \sum_{n=1}^{N} |x_n|^2 = \|\underline{x}\|_{\ell_2}^2$ (Induced norm)

<u>Def</u> $\underline{x}, \underline{y} \in \mathcal{X}$ are orthogonal with respect to the inner product $\langle \cdot, \cdot \rangle_{\mathcal{X}}$ if $\langle \underline{x}, \underline{y} \rangle_{\mathcal{X}} = 0$

$\Rightarrow$ Dot product $\underline{y}^H \underline{x} = 0$

<u>Cauchy-Schwartz Inequality</u>

$|y^H x| \le \|\underline{x}\|_{\ell_2} \|\underline{y}\|_{\ell_2}$ with equality iff $\underline{x} = \alpha \underline{y}$ for some $\alpha \in \mathbb{F}$

$|\cos \varphi| = \dfrac{\underline{y}^H x}{\|\underline{x}\|_{\ell_2} \|\underline{y}\|_{\ell_2}} \le 1$

Lemma

$\underline{x} \perp \underline{x}$ iff $\underline{x} = 0$

Proof (for $\mathbb{C}^N$ and dot products

$\Leftarrow$) If we assume $\underline{x} = 0 \Rightarrow \underline{x}^H x = \sum_{n=1}^{N} |x_n|^2 = 0$

$\Rightarrow$) If $\underline{x} \perp \underline{x} \Rightarrow x^H x = 0 \Rightarrow \|x\|_{\ell_2}^2 = 0 \Rightarrow \|x\|_{\ell_2} = 0 \Rightarrow x = 0$

QED